

## GENERALIZED KRUSKAL-WALLIS TESTS FOR COMPARING $k$ SAMPLES SUBJECT TO CENSORING

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*Summary:* Two generalizations of the Kruskal-Wallis k-sample test are proposed for censored data: one based on U-scores, as is Breslow's (1970) k-sample test, and the other based on Savage scores, as is Mantel's (1966) k-sample test. Contrary to Mantel's or Breslow's test, the proposed procedures are based on the permutation principle. However they rely on the assumption of no association between the censoring mechanism and the grouping variable, a situation usually found in clinical studies.

The development of the formulas is accomplished by redefining the Kruskal-Wallis test, using scores.

Monte Carlo experiments were conducted to investigate the behaviour of the tests under several situations in small samples. Finally an example is given and a computer program, facilitating the use of the test, is introduced.

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### 1. INTRODUCTION

A generalization of Wilcoxon's statistic for comparing two populations has been developed by Gehan (1965) for use when observations are subject to arbitrary one-sided censorship.

Similarly a generalization of Kendall's statistic for testing association between two continuous variables, one or both of them subject to arbitrary one-sided censorship has been proposed by Brown, Hollander and Korwar (1974). The primary motivation for this research was to obtain k-sample tests for censored data that are truly nonparametric, being based on the permutation principle, as are the two tests already mentioned.

Breslow (1974) has presented a "generalized Kruskal-Wallis test", but his test is not truly nonparametric, because permutational variances are only used for the components of the test statistic at "distinct failure times", not for the test statistic as a whole. Due to this fact, Breslow's test can be applied to situations, when k populations have been under study for different lengths of time, i.e. when censoring patterns differ between the groups. However, in randomized trials, in controlled clinical, therapeutical studies, where k populations are under study for equal lengths of time, the present test might be the most reliable one.

## 2. THE k-SAMPLE TEST BY KRUSKAL AND WALLIS

Suppose that k independent samples of sizes  $n_1, \dots, n_k$  are to be drawn from k univariate populations with unknown cumulative distribution functions  $F_1, F_2, \dots, F_k$ . Kruskal and Wallis (1952) discuss a test of the null hypothesis  $H_0: F_1 = F_2 = \dots = F_k$  against alternatives of the form  $F_c(x) = F(x - \theta_c)$ , with the  $\theta_c$ 's not all equal,  $c = 1, \dots, k$ , x being the variable under study.

Hajek and Sidak (1967) and Hajek (1969) have presented a general formulation of k-sample permutation tests, the test statistics being based on scores  $a_i$ ,  $i = 1, \dots, n$ , describing the relative position of the observations  $x_i$  in the combined sample of

size  $n$ .

$$(2.1) \quad Q = \left\{ \sum_{c=1}^k (S_c - E(S_c))^2 / n_c \right\} / \left[ \left\{ 1 / (n-1) \right\} \sum_{i=1}^n (a_i - \bar{a})^2 \right]$$

In equation (2.1)  $S_c$  denotes the sum of scores belonging to observations of the  $c$ -th sample of size  $n_c$ :  $S_c = \sum_{1 \in I_c} a_1$ .

It has been shown for a general class of scores  $a_i$  (cf. Hajek (1969)) that under  $H_0$   $Q$  asymptotically follows a  $\chi^2_{k-1}$ -distribution.

If the scores  $a_i$  are represented by the ranks of the observations, equation (2.1) specializes to

$$(2.2) \quad H = \left\{ 12 / (n(n+1)) \right\} \sum_{c=1}^k (R_c^2 / n_c) - 3(n+1)$$

or if ties of extent  $t$  occur, to

$$(2.3) \quad H = 12(n-1) \sum_{c=1}^k \left\{ \bar{R}_c - (n+1) / 2 \right\}^2 n_c / \left\{ n(n^2-1) - \sum_t t(t^2-1) \right\}.$$

This is Kruskal's  $H$  given by Kruskal (1952) and Kruskal and Wallis (1952), where  $R_c(\bar{R}_c)$  denotes the sum (mean) of ranks of sample  $c$ .

### 3. A REFORMULATION OF THE KRUSKAL-WALLIS TEST USING U-SCORES

As the Kruskal-Wallis test is based on ranks, we present a reformulation based on U-scores (Gehan, 1965) which allows generalizations in the case of censoring and define:

$$(3.1) \quad u_{ij} = \begin{cases} -\frac{1}{2} & \text{if } (x_i - x_j) \geq 0 \text{ and } \text{Min}(x_i, x_j) \text{ uncensored} \\ 0 & \text{otherwise} \end{cases} \quad i, j = 1, \dots, n$$

If variable  $x$  represents a survival time subject to arbitrary one-sided censoring, the sign of a difference  $(x_i - x_j)$  may be undefined because the smaller of  $x_i, x_j$  is censored.

If no censored observations occur, the rank of the  $p$ -th observation can be expressed by U-scores:

$$(3.2) \quad R_p = (n+1)/2 + \sum_{j=1}^n u_{pj}/2$$

Instead of using ranks we now attribute  $\sum_{j=1}^n u_{pj}$  to  $a_p$ , the score of the  $p$ -th observation.

As any of the scores  $u_{ij}$  used for calculation occur once as  $u_{ij}$  and once as  $u_{ji}$  and because by definition (3.1)  $u_{ij} = -u_{ji}$ , the mean of  $a_i$ ,  $\bar{a}$ , is 0.

Therefore, by insertion in equation (2.1), we get

$$(3.3) \quad H^B = (n-1) \left\{ \sum_{c=1}^k \left( \sum_{p \in I_c} \sum_{j=1}^n u_{pj} \right)^2 / n_c \right\} / \sum_{i=1}^n \left( \sum_{j=1}^n u_{ij} \right)^2$$

where  $I_c$  denotes the set of indices of observations belonging to the  $c$ -th sample. Because equation (3.3) defines a  $k$ -sample test for censored data based on the permutation principle and on U-scores,  $H^B$  exhibits a "Breslow-type" generalization of Kruskal's  $H$ . Breslow (1970) developed a  $k$ -sample test for censored data based on U-scores, but not on complete permutations of the test statistic.

#### 4. THE GENERALIZED KRUSKAL-WALLIS TEST BASED ON SAVAGE SCORES

One further advantage of reformulating the Kruskal-Wallis test using scores is the possibility of a straightforward generalization to several scoring systems. The previous chapters dealt

with the use of U-scores, where equal weight is given to each observation or individual. If equal weight is to be given to the time points where failures or deaths occur, the use of Savage (1956)-scores

$$(4.1) \quad 1_{ij} = u_{ij}/r_{ij}$$

as indicated. In (4.1)  $r_{ij}$  denotes the number of sampling units at risk to failure or death at the earlier of the two time points corresponding to indices  $i, j$ . Tests based on scores (4.1) have good power properties, if the underlying distributions are exponential, which has been confirmed by Peto and Peto (1972) for instance.

$H^M$ , Kruskal's "Mantel-type"  $H$ , was derived by insertion in equation (2.1) in a similar way to  $H^B$ , offering a complete permutational analogue to Mantel's  $k$ -sample test for censored data:

$$(4.2) \quad H^M = (n-1) \left\{ \sum_{c=1}^k \left( \sum_{p \in I_c} \sum_{j=1}^n 1_{pj} \right)^2 / n_c \right\} / \sum_{i=1}^n \left( \sum_{j=1}^n 1_{ij} \right)^2$$

## 5. A MONTE CARLO STUDY ON SIZE AND POWER OF THE TESTS

Monte Carlo experiments based on 1000 simulated trials of 3 groups each were conducted to investigate the behaviour of the two tests described. The simulation experiments were all carried out for expected overall percentages censored of 0, 40 and 80, following the methodology given in Lininger, Gail, Green and Byar (1979). Exponential failure times were assumed for all simulated trials.

A series of experiments was conducted for group sizes of 20 each. The means of the exponential failure time distributions for groups 1 and 2 are 1, the difference to the mean of group 3

is  $x$  (see Fig. 1). The power curves for an expected overall percentage censored of 80 were deleted from Fig. 1 for clarity; they never exceeded 0,25:

(a) exponential alternatives (b) translation alternatives

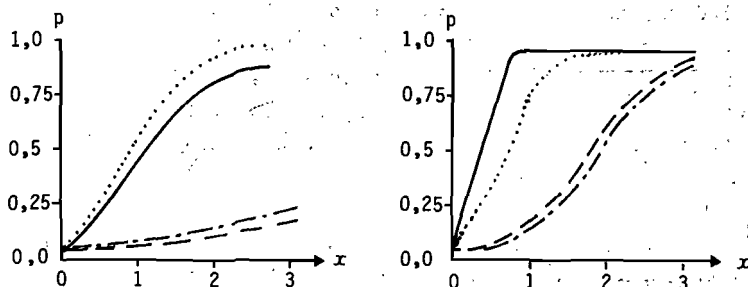


Fig. 1: Power curves for the tests in samples of size  $n_1 = n_2 = n_3 = 20$  from exponential distributions:

U - scores test: no censoring (—),  
40% censoring (---).  
Savage-scores test: no censoring (....),  
40% censoring (·-·-·)

$x$ ...difference between the means of groups 1 and 2 and the mean of group 3.

$p$ ...proportion where  $H_0$  is rejected at  $\alpha = 0,05$ .

Similar series of experiments were conducted for group sizes of 10 each, for group sizes of 6, 18 and 36 and for situations where the mean of group 2 was equally distant from the means of groups 1 and 3. The tests of these series exhibited slightly less power than in the series of Fig. 1, their size

was nearly nominal, as for the experiments in the series of Fig. 1. The size was even found to be nearly nominal in experiments for group sizes of 5 each.

The Monte Carlo experiments confirmed the Savage-scores tests to be more powerful if the underlying alternatives are exponential and the U-scores tests to be more powerful if the underlying alternatives are translations of exponential distributions. These results are in agreement with those derived by Lee, Desu and Gehan (1975) for two-sample tests.

## 6. A CALCULATED EXAMPLE AND FINAL DISCUSSION

Example: The survival times of 35 patients with a special disease, treated by 3 different therapies are:

Group 1: 1 2 3 3 5 7 9 10<sup>C</sup> 17<sup>C</sup> 18 18 20 23  
 Group 2: 2 8<sup>C</sup> 10 16 22<sup>C</sup> 24 27<sup>C</sup> 29 32 35<sup>C</sup> 38<sup>C</sup>  
 Group 3: 6 9 11 13<sup>C</sup> 15 25 26<sup>C</sup> 26<sup>C</sup> 28 36<sup>C</sup>

$x^C$  denoting a censored observation.

By means of a FORTRAN IV computer program called SUKRU (SURvival-KRUSkal), written by the author and available on request, both tests proposed by this paper were performed, yielding calculated significance probabilities of  $p = 0,019$  (using U-scores) and of  $p = 0,021$  (using Savage scores).

Due to the differences in the definitions of  $u_{ij}$  (3.1) and  $l_{ij}$  (4.1), procedures based on Savage scores are more sensitive to late events, giving equal weight to all time-points. Procedures based on U-scores are more sensitive to early events, giving less weight to late time-points when few individuals are still at risk. The construction of similar k-sample tests using other weighting functions of survival times or, identically, other scoring systems, is straightforward (e.g. for Tarone and Ware (1977)-type tests).

Our findings indicate that the proposed chi-square approximation to the distribution of the test statistic is quite accurate for group sizes of at least 5 or for a total of at least 4 groups with at least 3 observations per group, which agrees with limits given by Lienert (1973). If there is extensive tying or censoring, group sizes of at least 7 or 8 are recommended. Contrary to Mantel's or Breslow's test, the procedures proposed do not require the inversion of a covariance matrix, which simplifies computation.

In the case of two samples, Gehan's (1965) test is identical with the proposed generalization of the Kruskal-Wallis test based on U-scores.

Marcuson and Nordbrock (1981) presented a test which is identical to the generalized Kruskal-Wallis test based on U-scores, although their chi-square test statistic is based on a generalized inverse.

Mantel's and Breslow's k-sample tests can easily be applied to situations in which stratification seems appropriate; the same is true for the complete permutational analogues presented in this paper. An association of a patient's treatment and of the remaining life-span at the diagnosis of a cancerous disease could be analyzed by taking into account stratification factors such as histological type, site of the tumor or state of generalization of the disease. From an applied point of view, however, it is important to note that the number of stratification factors taken into account simultaneously is limited because of the rapidly decreasing sample sizes of subgroups or cells.

Stratified tests are performed by calculating

$$(5.1) \quad H^B = \sum_{c=1}^k \sum_{s=1}^S \left( \sum_{p \in I_{cs}} \sum_{j=1}^{n_s} u_{pj} \right)^2 (n_s - 1) / \left\{ n_{cs} \sum_{i=1}^{n_s} \left( \sum_{j=1}^{n_s} u_{ijs} \right)^2 \right\}$$

or similarly  $H^M$ , where  $S$  denotes the number of subgroups or cells defined by stratifying variables;  $n_s$ ,  $n_{cs}$ ,  $I_{cs}$  or  $u_{ijs}$  have the same meanings as  $n$ ,  $n_c$ ,  $I_c$  or  $u_{ij}$ , though for subgroup  $s$ . By way of conclusion, the proposed tests may represent a reliable alternative to Mantel's or Breslow's test, being based on the principle of permutation. If censoring does not occur, the redefinition of the test using scores, opening a class of Kruskal-Wallis type tests, is still of interest. However, if mechanisms of censoring or tying are not expected to be similar within the groups under comparison, use of complete permutations and the tests presented here may not be appropriate.

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#### REFERENCES

- BRESLOW, N.E. (1970). A generalized Kruskal-Wallis test for comparing  $K$  samples subject to unequal patterns of censorship. *Biometrika* 57, 579-594.
- BROWN, B.W., HOLLANDER, M. and KORWAR, R.M. (1974). Nonparametric tests of independence for censored data, with applications to heart transplant studies. Reliability and Biometry, F. Proschan, and R.J. Serfling (eds.) S.I.A.M., Philadelphia, 327-354.
- GEHAN, E.A. (1965). A generalized Wilcoxon test for comparing arbitrarily singly censored samples. *Biometrika* 52, 203-223.

- HAJEK, J. and SIDAK, Z. (1967). *Theory of Rank Tests*. Academia, Prague.
- HAJEK, J. (1969). *Nonparametric statistics*. Holden-Day, San Francisco.
- KRUSKAL, W.H. (1952). A nonparametric test for the several sample problem. *Ann. Math. Statist.* 23, 525-540.
- KRUSKAL, W.H. and WALLIS, W.A. (1952). Use of ranks in one-criterion variance analysis. *J. Amer. Statist. Assoc.* 47, 583-621.
- LEE, E.T., DESU, M.M. and GEHAN, E.A. (1975). A Monte Carlo study of the power of some two-sample tests. *Biometrika* 62, 425-431.
- LIENERT, G.A. (1973). *Verteilungsfreie Methoden in der Biostatistik*, Bd. 1, 2. Aufl., Anton Hain, Meisenheim am Glan, F.R.G.
- LININGER, L., GAIL, M.H., GREEN, S.B. and BYAR, D.P. (1979). Comparison of four tests for equality of survival curves in the presence of stratification and censoring. *Biometrika* 66, 419-428.
- MANTEL, N. (1966). Evaluation of survival data and two new rank order statistics arising in its consideration. *Cancer Chemotherapy Reports* 50, 163-170.
- MARCUSON, R. and NORDBROCK, E. (1981). A multivariate permutation test for the analysis of arbitrarily censored survival data. *Biometrical Journal* 23, 461-465.
- PETO, R. and PETO, J. (1972). Asymptotically efficient rank invariant test procedures (with discussion). *Journal of the Royal Statistical Society, Series A* 135, 185-206.

- SAVAGE, I.R. (1956). Contributions to the theory of rank order statistics - the two sample case. *Ann. Math. Statist.* 27, 590-615.
- TARONE, R.E. and WARE, J. (1977). On distribution-free tests for equality of survival distributions. *Biometrika* 64, 156-160.